

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (ME)

October 2017

ME-309 Design of Machine Elements-I

Date: 03.10.2017, Tuesday

Time: 1:00 p.m. To 4:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q – 1 (a) Attempt any Two. [08]

1. Which are the Consideration kept in mind when designing a friction clutch.
2. Write down the characteristics of material used for friction surface of a clutch.
3. Elaborate Slip and Creep of the belt?

Q – 1 (b) [05] A multiple disc clutch has five plates having four pairs of active friction surfaces. If the intensity of pressure is not exceed 0.127 N/mm^2 , find the power transmitted at 500 r.p.m. The outer and inner radii of friction surfaces are 125 mm and 75 mm respectively. Assume uniform wear and take coefficient of friction= 0.3.

OR

(b) [05] Write down the principles should be adopted to obtain maximum benefit from forged components.

Q – 2 (a) Attempt any Two. [10]

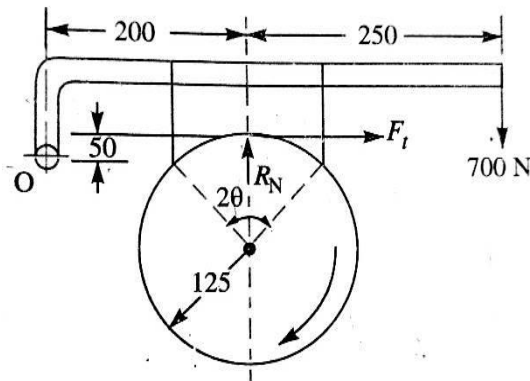
1. Write down the characteristics of material used for brake lining.
2. Describe the types of brakes in detail with their applications.
3. Derive the frictional torque equation for Disc clutch by considering uniform pressure condition.

Q – 2 (b) [05] A single block brake is shown in **fig.1**. The diameter of the drum is 250 mm and the angle of contact is 90° . If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake.

Q – 3 [07] Derive the equation for finding the length of open belt drive.

OR

Q – 3 [07] A belt 100 mm wide and 10 mm thick is transmitting power at 1000 meters/min. The net driving tension is 1.8 times the tension on the slack side. If the safe permissible stress on the belt section is 1.6 MPA, calculate the maximum power that can be transmitted at this speed. Assume density of the leather as 1000 kg/m^3 . Calculate the absolute maximum power that can be transmitted by this belt and the speed at which this can be transmitted.



All dimensions in mm.

Fig. 1

SECTION – II

Q - 4 Attempt any Two.

[08]

1. Derive an equation of strength for a sunk key with usual notations and sketch.
2. Determine an equation for cumulative damage in fatigue
3. Formulate an equation of required torque to raise the load,

Q – 5 (a) Design a clamp coupling to transmit 30 kW at 100 rpm. The allowable shear stress for the shaft and key of 40 MPa and the number of bolts connecting the two halves are six. The permissible tensile stress for the bolts is 70 MPa. The coefficient of friction between the muff and the shaft surface may be taken as 0.3. Assume width and thickness of the key are 22 mm and 14 mm respectively. [07]

(b) Explain the methods to reduce stress concentration in fatigue components. [08]

OR

(b) Explain the lame's equation used to determine wall thickness. [08]

Q – 6 Solve any two. [12]

(a) The cutter of a broaching machine is pulled by square threaded screw of 55 mm external diameter and 10 mm pitch. The operating nut takes the axial load of 400 N on a flat surface of 60 mm and 90 mm internal and external diameters respectively. If the coefficient of friction is 0.15 for all contact surfaces on the nut, determine the power required to rotate the operating nut when the cutting speed is 6 m/min. also find the efficiency of the screw.

(b) The layout of a transmission shaft carrying two pulley B and C and supported on bearing A and D is shown in **fig. 2** power is supplied to the shaft by means of a vertical belt on pulley B, which is then transmitted to pulley C carrying a horizontal belt. The maximum tension in the belt on pulley B is 2.5 kN. The angle of wrap for both the pulley is 180° and the coefficient of friction is 0.24. The shaft is made of plain carbon steel having shear stress is 67 N/mm^2 . Determine the shaft diameter by equivalent torsional moment.

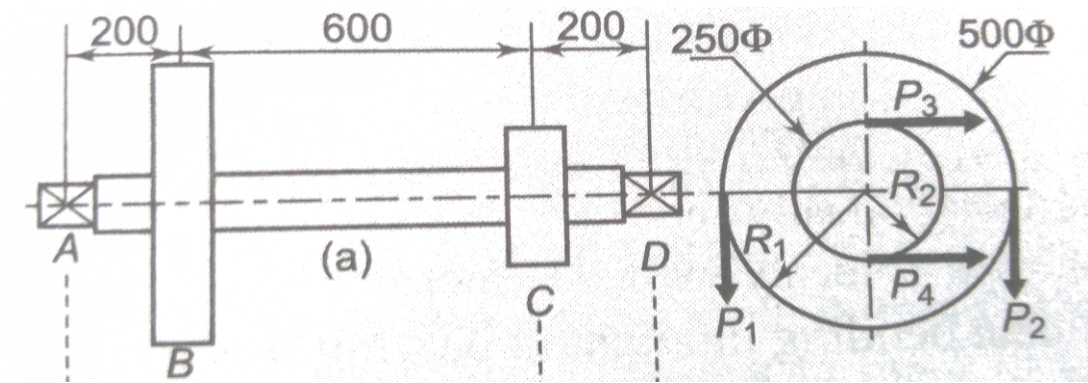


Fig. 2

- (c) A stepped steel shaft ABCDE supported in bearings as shown in **fig. 3** is supported to a non-rotating force of 4000 N. shaft is rotating and machined from steel with an ultimate tensile strength of 500 MPa. Taking surface finish factor as 0.8, size factor as 0.85, reliability factor 0.897. Find the expected life of the shaft, if the theoretical stress concentration factor as 1.68 and notch sensitivity as 0.78 under these condition.

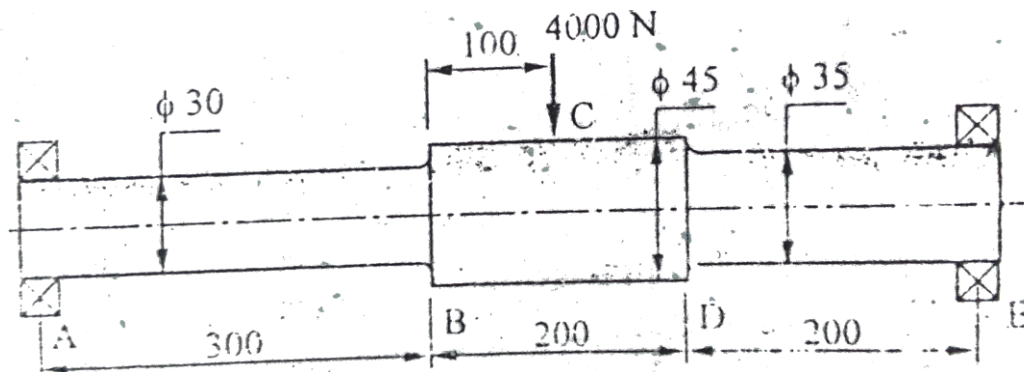


Fig. 3
